

Static Analysis and Verification

Automata-based LTL Model Checking

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Introduction

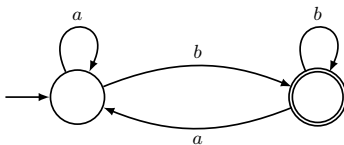
- We need to check whether $M \models \varphi$ holds for a Kripke structure M and an LTL formula φ .
- We are going to use an automata-theoretic approach to solve the above problem.
- The semantics of LTL formulae is defined over infinite paths—hence, when considering labelling of states as letters, we need to work with infinite words over the alphabet 2^{AP} .
- We need a suitable kind of automata to represent languages of infinite words: we are going to use the so-called Büchi automata (BA) and their variants (called, in general, ω -automata).
 - ▶ At the first glance, Büchi automata look like ordinary finite automata, but they accept infinite words by looping through accepting states.
 - ▶ We transform a Kripke structure M to a BA $\mathcal{B}_M$ accepting words that correspond to the paths in $\bigcup_{s_0 \in S_0} \Pi(M, s_0)$ when only the labelling of the states is considered.
 - ▶ We translate an LTL formula φ into a BA $\mathcal{B}_{\neg\varphi}$ accepting words corresponding to paths π such that $\pi \not\models \varphi$. (We do not refer to any concrete M and consider paths in all Kripke structures.)
 - ▶ We check that $L(\mathcal{B}_M) \cap L(\mathcal{B}_{\neg\varphi}) = \emptyset$.

Büchi Automata for use in LTL Model Checking

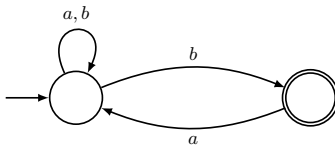
Büchi Automata

- A (non-deterministic) Büchi automaton $\mathcal{B}$ is a tuple $\mathcal{B} = (Q, \Sigma, \delta, Q_0, F)$ where
 - ▶ Q is a finite set of states,
 - ▶ Σ is a finite alphabet,
 - ▶ $\delta \subseteq Q \times \Sigma \times Q$ is the transition relation,
 - ▶ $Q_0 \subseteq Q$ is the set of initial states,
 - ▶ $F \subseteq Q$ is the set of accepting states.
- Provided $(q_1, a, q_2) \in \delta$, we often write $q_1 \xrightarrow{a} q_2$.
- The language of a BA $\mathcal{B} = (Q, \Sigma, \delta, Q_0, F)$ is defined as follows:
 - ▶ A run ϱ of $\mathcal{B}$ over an infinite word $w = a_0a_1a_2 \dots \in \Sigma^\omega$ is an infinite sequence $q_0q_1q_2 \dots \in Q^\omega$ of states such that $q_0 \in Q_0$ and $\forall i: q_i \xrightarrow{a_i} q_{i+1}$.
 - ▶ A run ϱ is accepting iff $\text{inf}(\varrho) \cap F \neq \emptyset$ where $\text{inf}(\varrho)$ is the set of states that appear infinitely often in ϱ .
 - ▶ The language of $\mathcal{B}$ is defined as $L(\mathcal{B}) = \{w \in \Sigma^\omega \mid \text{there is an accepting run of } \mathcal{B} \text{ over } w\}$.

Two Examples of BAs



A BA accepting the language of infinite words over $\Sigma = \{a, b\}$ in which *b* appears infinitely often.



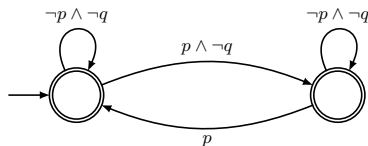
A BA accepting the language of infinite words over $\Sigma = \{a, b\}$ in which *ba* appears infinitely often.

ω -Regular Languages

- When dealing with BA with $\Sigma = 2^{AP}$, we often describe (sets of) transitions using **propositional formulae**:

- ▶ e.g., for $AP = \{p, q, r\}$, we may write a transition labelled with $p \wedge \neg q$ instead of two transitions labelled with $\{p\}$ and $\{p, r\}$.

- An example:



the BA describes the set of words over 2^{AP} , $AP = \{p, q, r\}$, such that q may appear (if at all) at *even occurrences* of p only.

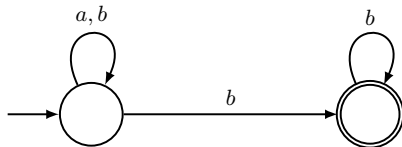
- The above language is **not expressible using LTL**.
 - ▶ BA have a **strictly higher expressive power** than LTL.
 - ▶ The languages that are accepted by some BA are called **ω -regular**.

Alternative Accepting Conditions

- Several other forms of accepting conditions replacing the simple set of accepting states F are used:
 - ▶ **generalized Büchi**: $\mathcal{F} \subseteq 2^Q$ —a run ϱ is accepting iff $\forall F \in \mathcal{F}: \inf(\varrho) \cap F \neq \emptyset$.
 - ▶ **Muller**: $\mathcal{F} \subseteq 2^Q$ —a run ϱ is accepting iff $\exists F \in \mathcal{F}: \inf(\varrho) = F$.
 - ▶ **Streett**: $\mathcal{F} \subseteq 2^Q \times 2^Q$ —a run ϱ is accepting iff $\forall (E, F) \in \mathcal{F}: \inf(\varrho) \cap E \neq \emptyset \implies \inf(\varrho) \cap F \neq \emptyset$.
 - ▶ **Rabin**: $\mathcal{F} \subseteq 2^Q \times 2^Q$ —a run ϱ is accepting iff $\exists (E, F) \in \mathcal{F}: \inf(\varrho) \cap E = \emptyset \wedge \inf(\varrho) \cap F \neq \emptyset$.
 - ▶ **parity**: states of $\mathcal{B}$ are labelled with colours from the set $C = \{0, \dots, k\}$ by a function $c: Q \rightarrow C$. A run ϱ is accepting iff $\min\{c(q) \mid q \in \inf(\varrho)\}$ is even (alternative definitions for max/odd).
 - ▶ **Emerson-Lei**: states of $\mathcal{B}$ are labelled with sets of colours from C and the acceptance condition is given by an arbitrary Boolean formula φ over atoms of the form $\inf(c_i)$ for $c_i \in C$. A run ϱ is accepting iff $\inf(\varrho)$ is a model of φ .
 - ▶ **transition-based acceptance**: as above, but states are substituted with transitions.
- All the above conditions yield automata of equal expressive power.

Deterministic BAs

- When considering the basic Büchi (or generalized Büchi) acceptance condition, **deterministic BAs** are **strictly less powerful** than ordinary (non-deterministic) BAs.



- The above BA expressing the language of words over $\Sigma = \{a, b\}$ in which **eventually only b appears** (i.e., $(a + b)^* b^\omega$) does not have a deterministic variant.
- Deterministic and non-deterministic **Muller, Streett, Rabin, parity, and Emerson-Lei automata** have the **same expressive power**.

Complementation of BAs

- The automata-theoretic approach to LTL model checking could be formulated as checking whether $L(\mathcal{B}_M) \subseteq L(\mathcal{B}_\varphi)$, which would naturally reduce to using complementation to check $L(\mathcal{B}_M) \subseteq L(\mathcal{B}_\varphi)$ as $L(\mathcal{B}_M) \cap \overline{L(\mathcal{B}_\varphi)} = \emptyset$.
- Due to the non-equivalent power of deterministic and non-deterministic BA, complementation is much more complicated than for finite-word finite automata.
- However, BAs are still closed wrt complementation:
 - ▶ One can complement BAs, e.g., using the so-called Safra construction going through deterministic Rabin automata.
 - The complement of a BA with n states using this way has $2^{\mathcal{O}(n \log(n))}$ states (more precisely, $12^n n^{2n}$ for Safra (into Rabin) and $2n^n n!$ for Piterman (into parity)).
 - ▶ There are other procedures for complementation (the lower bound is $\Omega(\frac{1}{n}(0.76n)^n)$):
 - Ramsey-based, determinization-based, rank-based (tight: $\mathcal{O}(n(0.76n)^n)$), slice-based, learning-based, subset-tuple construction, semideterministic-based, decomposition-based (+ specialized procedures for subclasses).
- To avoid the complex complementation of BAs, complementation is usually done on the level of formulae, and the model checking checks that $L(\mathcal{B}_M) \cap L(\mathcal{B}_{\neg\varphi}) = \emptyset$.

Emptiness of BAs

- Emptiness of a given BA $\mathcal{B}$ can be checked in the following way:
 - ▶ compute the SCCs of $\mathcal{B}$, which can be done using Tarjan's algorithm in time linear in the size of $\mathcal{B}$,
 - ▶ check whether there is a non-trivial SCC that contains an accepting state and is reachable from some initial state.
- The above procedure can be done in time $\mathcal{O}(|Q| + |\delta|)$.
- Nested depth-first search—two interleaved depth-first searches:
 - ▶ The outer DFS searches for accepting states and the inner DFS tries to find a loop on the encountered, fully-expanded by the outer DFS, accepting states (while going through states not visited by the inner DFS).
 - ▶ Note: A naive two-phase DFS (first find accepting states, then search from each of them for a loop) gives time complexity $\mathcal{O}(|Q| \cdot (|Q| + |\delta|))$.
- *In the literature, various improved versions of both the SCC-based as well as the nested DFS have been proposed: these are beyond the scope of this lecture.*

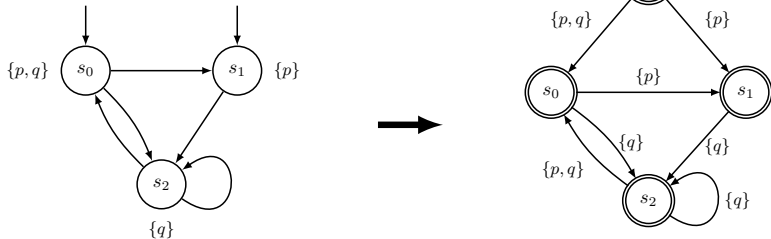
Product of BAs

- Given two BAs $\mathcal{B}_1, \mathcal{B}_2$, constructing a BA accepting the language $L(\mathcal{B}_1) \cap L(\mathcal{B}_2)$ is easy.
- However, one has to be careful of the fact that accepting states may be reached in $\mathcal{B}_1$ and $\mathcal{B}_2$ at different times.
 - ▶ Have two copies of the cross product of the transition graphs of $\mathcal{B}_1$ and $\mathcal{B}_2$.
 - ▶ For $q_1^1 \in F_1$, redirect each transition going from a state (q_1^1, q_1^2) to (q_2^1, q_2^2) in the first copy of the cross product to go from (q_1^1, q_1^2) in the first copy to (q_2^1, q_2^2) in the second copy.
 - ▶ Redirect in a similar fashion transitions from the second copy back to the first one.
 - ▶ Consider as accepting the states (q_1, q_2) of the second copy where $q_2 \in F_2$.
- In practice, we often use an algorithm that outputs a generalized BA.
- In the LTL model checking procedure, the construction of the product may be simplified since $\mathcal{B}_M$ for a Kripke structure M will have all states accepting:
 - ▶ Hence, no need to create two copies of the cross product.
 - ▶ One can consider as accepting the states of the cross product in which the $\mathcal{B}_{\neg\varphi}$ component reaches an accepting state.

From Kripke Structures to Büchi Automata

From KS to BA

- We transform a given Kripke structure $M = (S, S_0, R, L)$ over atomic propositions from AP to the Büchi automaton $\mathcal{B}_M = (S \cup \{q_0\}, 2^{AP}, \delta, \{q_0\}, S \cup \{q_0\})$ where
 - ▶ $q_0 \notin S$ and
 - ▶ δ is the smallest relation such that
 - if $(s_1, s_2) \in R$, then $(s_1, L(s_2), s_2) \in \delta$ and
 - if $s_0 \in S_0$, then $(q_0, L(s_0), s_0) \in \delta$.
- We have that $L(\mathcal{B}_M) = \{L(s_0)L(s_1)L(s_2)\dots \mid s_0 \in S_0 \wedge s_0s_1s_2\dots \in \Pi(M, s_0)\}$.
- An example:



From LTL Formulae to Büchi Automata

The Idea of Going from LTL to BA

- We consider the **basic connectives** ($\neg$, $\vee$, X , U) only and we skip the use of the implicit A path quantifier at the beginning of the formulae.
- We introduce a **state** q for each **consistent subset of the set of subformulae** of the given formula and **their negations**: these are assumed to hold in q .
- We add **transitions** according to the **observed changes in the validity of atomic propositions** (the sets of the new valid atomic propositions will label the transitions) and according to the **temporal operators** that appear in the formulae present in the states.
- We use **generalized BA**: **one accepting condition for each until**.
 - ▶ The generalized BA may be **converted to plain BA** in a similar way as in the product construction (just using as many copies as the number of accepting conditions is).
- Various **alternative, more optimised constructions** have been studied (and are available in tools such as `ltl2ba`, `ltl2tgba` ([Spot](#)), `ltl3tela`, `Rabinizer`, ...).

The FL Closure of a Formula

- Let φ be an LTL formula built over atomic propositions from AP using the connectives $\neg$, $\vee$, X , and U . The **Fischer-Ladner (FL) closure** $cl(\varphi)$ of φ is defined inductively on the structure of φ (assuming that $\neg\neg\varphi \equiv \varphi$):

- ▶ $cl(p) = \{p, \neg p\}$ for $p \in AP$,
- ▶ $cl(\neg\varphi) = cl(\varphi) \cup \{\neg\varphi\}$,
- ▶ $cl(\varphi_1 \vee \varphi_2) = cl(\varphi_1) \cup cl(\varphi_2) \cup \{\varphi_1 \vee \varphi_2, \neg(\varphi_1 \vee \varphi_2)\}$,
- ▶ $cl(X\varphi) = cl(\varphi) \cup \{X\varphi, \neg X\varphi\}$,
- ▶ $cl(\varphi_1 U \varphi_2) = cl(\varphi_1) \cup cl(\varphi_2) \cup \{\varphi_1 U \varphi_2, \neg(\varphi_1 U \varphi_2)\}$.

- Example:

$$cl(pUq \vee (\neg p)Uq) = \left\{ \begin{array}{ll} pUq \vee (\neg p)Uq, & \neg(pUq \vee (\neg p)Uq), \\ pUq, & \neg(pUq), \\ (\neg p)Uq, & \neg((\neg p)Uq), \\ p, \neg p, & q, \neg q \end{array} \right\}$$

Consistent Sets of Formulae

- We want to restrict the construction to sets of formulae that **do not contain contradictory formulae** (i.e., formulae that can never hold together).
- Given an LTL formula φ with the chosen basic connectives, we call a set $q \subseteq cl(\varphi)$ **consistent** iff the following conditions hold:
 - 1 $\forall \psi \in cl(\varphi): \psi \in q \iff \neg\psi \notin q.$
 - 2 $\forall (\psi_1 \vee \psi_2) \in cl(\varphi): (\psi_1 \vee \psi_2) \in q \iff \psi_1 \in q \vee \psi_2 \in q.$
 - 3 $\forall (\psi_1 U \psi_2) \in cl(\varphi): \psi_2 \in q \implies (\psi_1 U \psi_2) \in q.$
 - 4 $\forall (\psi_1 U \psi_2) \in cl(\varphi): (\psi_1 U \psi_2) \in q \wedge \psi_2 \notin q \implies \psi_1 \in q.$

Constructing $\mathcal{B}_\varphi$

- Given an LTL formula φ built over atomic propositions from AP using the basic connectives $\neg$, $\vee$, X , U , the **generalized BA** $\mathcal{B}_\varphi = (Q, 2^{AP}, \delta, Q_0, \mathcal{F})$ is defined as follows:
 - ▶ $Q = \{q_0\} \cup \{q \subseteq cl(\varphi) \mid q \text{ is consistent}\}$, $q_0 \notin 2^{cl(\varphi)}$, and $Q_0 = \{q_0\}$.
 - ▶ $\delta \subseteq Q \times 2^{AP} \times Q$ satisfies the following conditions — $(q_0, a, q) \in \delta$ iff
 - 1 $q \neq q_0$,
 - 2 $\varphi \in q$, and
 - 3 $a = q \cap AP$;
 - ▶ and $(q_1, a, q_2) \in \delta$ for $q_1 \neq q_0$ iff
 - 1 $q_2 \neq q_0$,
 - 2 $a = q_2 \cap AP$,
 - 3 $\forall (X\psi) \in cl(\varphi): (X\psi) \in q_1 \iff \psi \in q_2$,
 - 4 $\forall (\psi_1 U \psi_2) \in cl(\varphi): (\psi_1 U \psi_2) \in q_1 \wedge \psi_2 \notin q_1 \implies (\psi_1 U \psi_2) \in q_2$,
 - 5 $\forall (\psi_1 U \psi_2) \in cl(\varphi): (\psi_1 U \psi_2) \notin q_1 \wedge \psi_1 \in q_1 \implies (\psi_1 U \psi_2) \notin q_2$.
 - ▶ $\mathcal{F} = \{\{q \in Q \setminus \{q_0\} \mid \psi_2 \in q \vee (\psi_1 U \psi_2) \notin q\} \mid (\psi_1 U \psi_2) \in cl(\varphi)\}$.
 - Guarantees that each until (once encountered) will reach its end (i.e., a state where its right operand holds).
- We have that $L(\mathcal{B}_\varphi) = \{L(s_0)L(s_1)L(s_2)\dots \mid \text{there is a KS } M = (S, S_0, R, L) \text{ over } AP \text{ such that } s_0 \in S_0, s_0s_1s_2\dots \in \Pi(M, s_0), \text{ and } M, s_0s_1s_2\dots \models \varphi\}$.

An Example of Translating from LTL to BA

■ Consider $\varphi = pUq$:

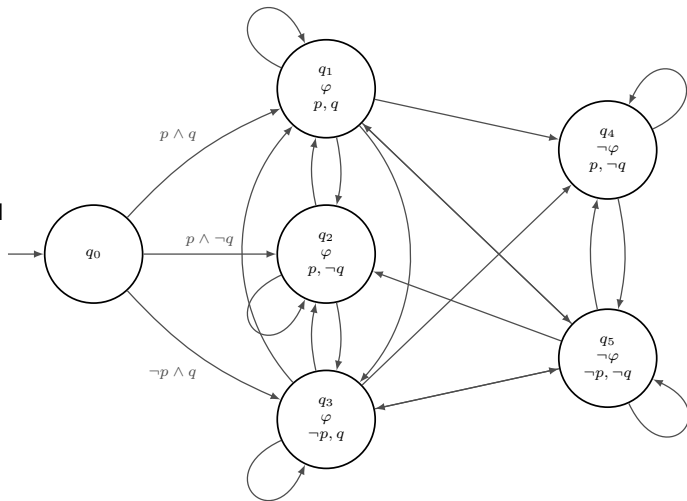
▶ $cl(\varphi) = \{p, \neg p, q, \neg q, \varphi, \neg\varphi\}$.

▶ Consistent subsets of $cl(\varphi)$:

- $q_1 = \{\varphi, p, q\}$,
- $q_2 = \{\varphi, p, \neg q\}$,
- $q_3 = \{\varphi, \neg p, q\}$,
- $q_4 = \{\neg\varphi, p, \neg q\}$,
- $q_5 = \{\neg\varphi, \neg p, \neg q\}$.

▶ $\mathcal{B}_\varphi$ is shown on the right (not all labels are shown).

▶ $\mathcal{F} = \{\{q_1, q_3, q_4, q_5\}\}$.



The Top Level of the LTL Model Checking Algorithm

A Naive LTL Model Checking Algorithm

■ A naïve procedure:

- 1 generate the **KS** M for the given system to be verified and the atomic observations AP of interest,
- 2 translate M to the **BA** $\mathcal{B}_M$,
- 3 **negate** the given LTL formula φ to be checked and translate the negation into the **BA** $\mathcal{B}_{\neg\varphi}$,
- 4 construct the product **BA** $\mathcal{B}_M \times \mathcal{B}_{\neg\varphi}$ representing the language $L(\mathcal{B}_M) \cap L(\mathcal{B}_{\neg\varphi})$,
- 5 check **language emptiness** of $\mathcal{B}_M \times \mathcal{B}_{\neg\varphi}$:
 - if $L(\mathcal{B}_M \times \mathcal{B}_{\neg\varphi})$ is **empty**, φ **holds** for the given system,
 - otherwise return a path corresponding to some element from the intersection as a **counterexample** to the property being checked.

On-the-Fly LTL MC Algorithm

- Differences of **on-the-fly model checking** from the **naïve procedure**:
 - ▶ Do not generate the KS M and the BA $\mathcal{B}_M$ first, only then constructing the product with the negated property BA, followed by checking its emptiness.
 - ▶ Instead, construct $\mathcal{B}_{\neg\varphi}$ and use it to **control the construction** of $\mathcal{B}_M$ and the product $\mathcal{B}_M \times \mathcal{B}_{\neg\varphi}$ while **continuously checking for accepting loops**:
 - if an accepting loop is detected, **immediately stop** and **print out a counterexample** without generating further states (faulty systems tend to have many strange states due to not obeying the intended invariants),
 - when some transition from the state of $\mathcal{B}_M$ that is currently being explored **cannot be composed** with the currently executable transitions of $\mathcal{B}_{\neg\varphi}$, **do not follow it** (no counterexample can be reached via the transition—hence, the sub-state space reachable (exclusively) via it needs not be explored).
 - ▶ Combine the on-the-fly generation of states of M with suitable **state space reduction techniques**, e.g.,
 - **partial order reduction** (exploring only some interleavings of the concurrent processes running in the verified system) or
 - **symmetry reduction** (do not explore states that are indistinguishable from some already generated states wrt the property being checked),
 - **bit-state hashing** (do not distinguish states with the same hash), ...